



MAÏWEN DEMEULLE

Master's Graduate in Bio-Informatics and Biostatistics

@maïwen.demeulle@gmail.com
prodstation1.myds.me:8443/

+33 7 82 40 65 70
maïwen-demeulle

IDF, FRANCE
MaiDeme

EDUCATION

Master in Bio-Informatics and Biostatistics

Université Paris-Saclay

Sept 2023 – August 2025

Structural Bioinformatics Statistics and Machine Learning

NGS Metagenomics Comparative Genomics

- Master's degree in Bioinformatics emphasizing genomic, biomedical data analysis algorithms and artificial intelligence. *Highest honors (16.2/20).*

Double Bachelor in Mathematics and Life Science

Université Paris-Saclay

Sept 2020 – June 2023

Algebra Probability theory Calculus Genetics

Statistical Inference Differential Equations

Molecular biology Biochemistry Microbiology

- Three-year Double Degree Bachelor's in Mathematics and Life Sciences, providing fundamental knowledge in biology and mathematics and focusing on interdisciplinary project. *Top student.*

EXPERIENCE

M2 Internship: Resolution of the pan-algal arctic specific genome

CQSB → HEAL Team

March – August 2025

IBPS-Paris

Snakemake Nanopore Singularity Evolution

Genome Assembly Genome Annotation Docker

- Creating a pipeline to assemble and annotated Algae genome from long, short reads and RNA-seq.
- Performing comparative analysis of a collection of genomes including the newly assembled genomes.

M1 Internship : Computational Biology

Groupe Roullier → Centre Mondial de l'Innovation

May – Aug 2024

Saint-Malo, FRANCE

R Shiny Python Virtual Screening

Molecular Docking

SKILLS

C/C++ Python R/R Shiny Bash

Java PostgreSQL Slurm Git

LaTeX Django Docker Snakemake

LANGUAGES

French Native

English C1: Linguaskill

German

Italian

PROJECTS

Phylogeny project

First year master

Reconstruction of the species tree among several mammals from DNA sequences using Needleman-Wunsch algorithm, as well as NJ and UPGMA, for tree construction. Code was written in C.

Annotathon

First year master

Bioinformatics training through metagenomic sequence annotation on the Annotathon website.

Java project

First year master

Recreated the classic Pac-Man video game in Java as part of a project.

Machine Learning Challenge

Second year master

- Extensive literature review on virtual screening tools from drug design, aiming to adapt these methods for molecules of interest in agriculture.
- Developping a Shiny application to visualize data from wet lab experiments.

Part time Internship: Assessment of the utility of synthetic genome for population genetics.

LISN → BioInfo Team

📅 Sept – Dec 2023

📍 Saclay, FRANCE

Python

Machine Learning

Artificial genome

- Local ancestry inference tasks on synthetics data from a Generative Adversarial Network (GAN) model.
- Labelling data using a k-nearest neighbour (knn) model.

Bachelor Internship

IMO (Mathematical Institute of Orsay)

📅 April – May 2022

📍 Orsay, FRANCE

Leslie population age distribution model

PageRank

- Working on the Perron-Frobenius theorem and some of its applications (PageRank, Leslie matrix).

PUBLICATIONS

📄 Journal Articles

- A. Szatkownik, L. Planche, **M. Demeulle**, *et al.*, "Diffusion-based artificial genomes and their usefulness for local ancestry inference," *bioRxiv*, 2024. DOI: 10.1101/2024.10.28.620648.

Participated in a classification challenge as part of a machine learning course.

Repro-Hackathon

📅 Second year master

Reproduced RNA-Seq analysis from article using Docker and Snakemake to ensure reproducibility and automation of workflows

Web Project

📅 Second year master

Created a website using Django to explore genomic data, perform sequence-based searches in a database and contribute to a collaborative genome annotation.

Game Jam

📅 January 2025

Participated in a 48-hour game jam, contributing to game design, development, and testing.

ACTIVITIES

Guitar and piano (self-taught)

Archery

Video Games

Tutor for M1 students

Webmaster of a student association

REFERENCES

Mr. Louis Graf

@ Internship Supervisor, Sorbonne University

✉ louis.graf@sorbonne-universite.fr